

Shaik Mahammad Khaleel

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SUMMARY

Hands on experience in synthesis, STA, Floor Planning, Power Planning, Placement, CTS, Routing and IR Drop Analysis.

Experience in lower technology nodes such as TSMC 16nm and 28nm and able to handle complex designs with multimillion gates and tens of macros.

Having better understanding of clock tree structure and implementation.

Hands on experience in low power implementation techniques such as ON-OFF and Multi-Voltage designs.

Have solid experience in STA, timing Eco's, functional Eco's, metal Eco's implementation.

Have hands on experience in DRC, LVS, LEC, DRV fixes, Crosstalk Analysis, Glitches, Antenna and IR/EM fixes.

Proficient in scripting languages TCL, PERL and good at UNIX.

WORK EXPERIENCE

3 years of experience in VLSI Physical design at Thinsil technologies Pvt Ltd.

Handled two projects and 3 blocks.

ACADEMIC QUALIFICATIONS

Qualification	Specialization	University/Board	Institute	Year	CGPA/Percentage
B.Tech	EEE	JNTUK	Tirumala Engineering college	2020	8.34 CGPA
Diploma	EEE	AP SBTET	A.M. Reddy Memorial college	2017	85.27%
Matriculation	SSC	Board of Secondary Education Andhra Pradesh	B R C high school	2014	9.2 CGPA

TOOLS

Synthesis	Design Compiler, Genus and FC shell
Place and Route	ICC2, FC shell and Innovus
Timing Analysis	Prime Time and Tempus
Sign off	Voltus, Quantus and calibre

PROJECT DEATILS

PROJECT I : Biltwell-v2

Client : Brite Semiconductors

Instance Count : 0.2M

Number of Macros : 42

Technology node : TSMC 28nm

Metal Layers : 9

Number of Clocks : 6

Operating Frequency : 480 MHz

Tools used : ICC2 and PT

Challenges faced : In this project, I was responsible for doing Floor Plan to Routing and fixing remaining timing violations. Timing closure is critical and found out Module splitting was the reason, applied regions, tried various experiments around it and resolved timing issues. Finally, did Physical Verification checks and fixed the violations.

PROJECT II : FALCON 7nm (Block 8 and Block 10)

Instance Count : 0.32 M and 0.8 M

Number of Macros : 0/25

Technology node : TSMC 7nm

Operating Frequency : 680 MHz

Number of Clocks : 4/3 (8 Generated clocks)

Tools used : Genus, Innovus, Tempus, Quantus, Voltus, calibre

Challenges faced : Block level handling including synthesis, PD, STA and PV closure

DECLARATION

I do hereby declare that the above particulars of facts and information stated are true, correct, and complete to the best of my belief and knowledge.

Place : Bangalore

Signature

Date : 03 Feb, 2024

Shaik Mahammad Khaleel